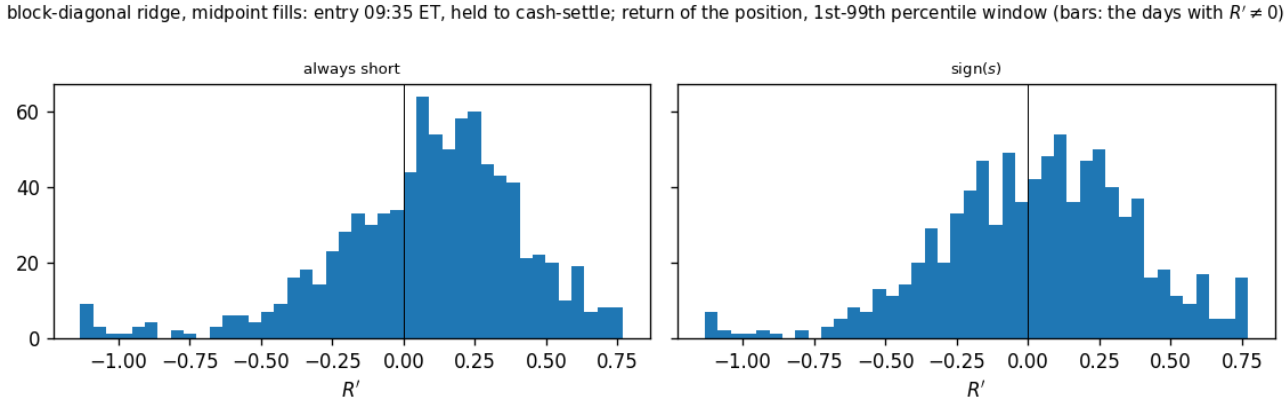
Rule table by strategy — entry 09:35 ET, held to cash-settle

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	863	0.076	0.385	-3.97	0.90	-2.97	23.37	5.76	3.11	0	0.0	2.44
sign(s) (± 1 *on the sign of* s)												
baseline (HAR + calendar OLS)	863	0.051	0.390	-3.47	3.97	-0.09	20.21	3.83	2.07	232	26.9	1.40
block-diagonal ridge	863	0.033	0.391	-3.47	3.97	-0.08	19.80	2.46	1.33	306	35.5	0.67
block-diagonal ridge, without the FOMC columns	863	0.033	0.391	-3.47	3.97	-0.08	19.80	2.48	1.34	307	35.6	0.68
LightGBM†	863	0.052	0.389	-3.47	3.97	-0.06	20.24	3.96	2.14	235	27.2	1.47
XGBoost†	863	0.051	0.389	-3.47	3.97	-0.09	20.22	3.86	2.08	231	26.8	1.42
lasso (causally tuned)†	863	0.041	0.391	-3.47	3.97	-0.01	19.94	3.10	1.67	283	32.8	1.01
lasso ($\alpha = 10^{-4}$)	863	0.028	0.392	-3.47	3.97	-0.06	19.70	2.07	1.12	313	36.3	0.46
elastic net (causally tuned)†	863	0.039	0.391	-3.47	3.97	-0.10	19.92	2.92	1.58	283	32.8	0.91

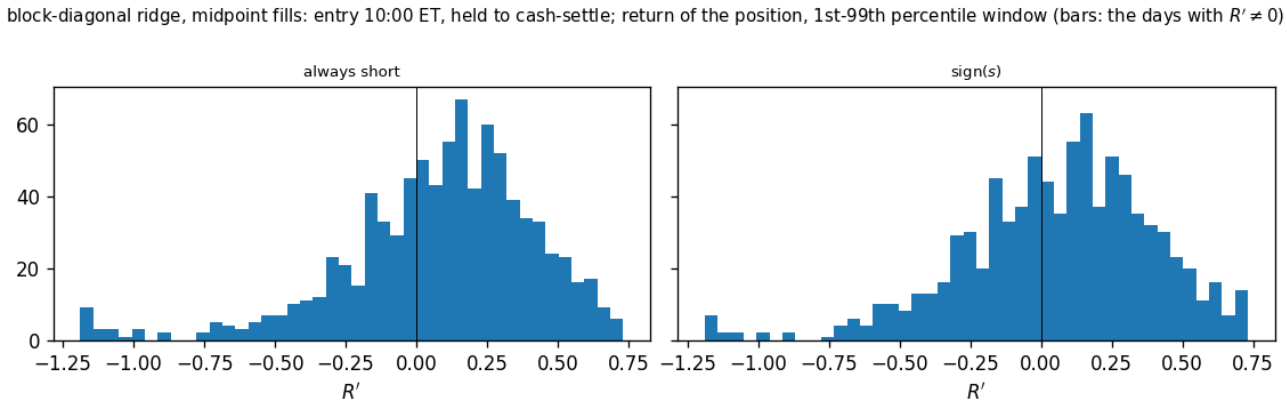


Distribution of the return of the position taken at 09:35 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 09:35 ET, hold those strikes to the official close, and delta-hedge at 10:00 and then every 30 minutes. 863 expiration days (the other 3 of the deck's 866 days have no quoted straddle at this stamp), 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread). **The 09:35 entry.** The first stamp of the session is 09:35, and no forecast bar starts there: the forecasts are for half-hour bars 09:30–10:00, 10:00–10:30, and so on. The forecast used at 09:35 is therefore the one issued at 09:30 for the 09:30–10:00 bar, which is known five minutes before the entry. It covers 09:30–10:00 while the straddle is entered at 09:35; it is used as issued, with no rescaling for the five minutes. The implied side is priced the same way: its diurnal share is the 09:30–10:00 bar's share of the rest of the session, averaged over prior days only (the same expanding mean as at every other clock), and the hours to the close are the chain's, 6.42 hours (6 h 25 min). This design was fixed before the 09:35 numbers were computed.

Rule table by strategy — entry 10:00 ET, held to cash-settle

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	864	0.078	0.371	-2.95	0.89	-1.96	9.38	6.15	3.32	0	0.0	2.29
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	864	0.045	0.377	-2.95	1.39	-1.41	7.97	3.51	1.90	207	24.0	0.89
block-diagonal ridge	864	0.058	0.375	-2.95	1.39	-1.43	8.32	4.52	2.44	219	25.3	1.44
block-diagonal ridge, without the FOMC columns	864	0.054	0.375	-2.95	1.39	-1.40	8.20	4.23	2.28	221	25.6	1.28
LightGBM†	864	0.056	0.375	-2.95	1.39	-1.42	8.25	4.36	2.35	199	23.0	1.34
XGBoost†	864	0.052	0.376	-2.95	1.39	-1.40	8.15	4.08	2.20	190	22.0	1.20
lasso (causally tuned)†	864	0.052	0.376	-2.95	1.39	-1.38	8.13	4.05	2.19	205	23.7	1.18
lasso ($\alpha = 10^{-4}$)	864	0.059	0.375	-2.95	1.39	-1.43	8.34	4.60	2.48	208	24.1	1.48
elastic net (causally tuned)†	864	0.048	0.376	-2.95	1.39	-1.35	8.03	3.78	2.04	210	24.3	1.03

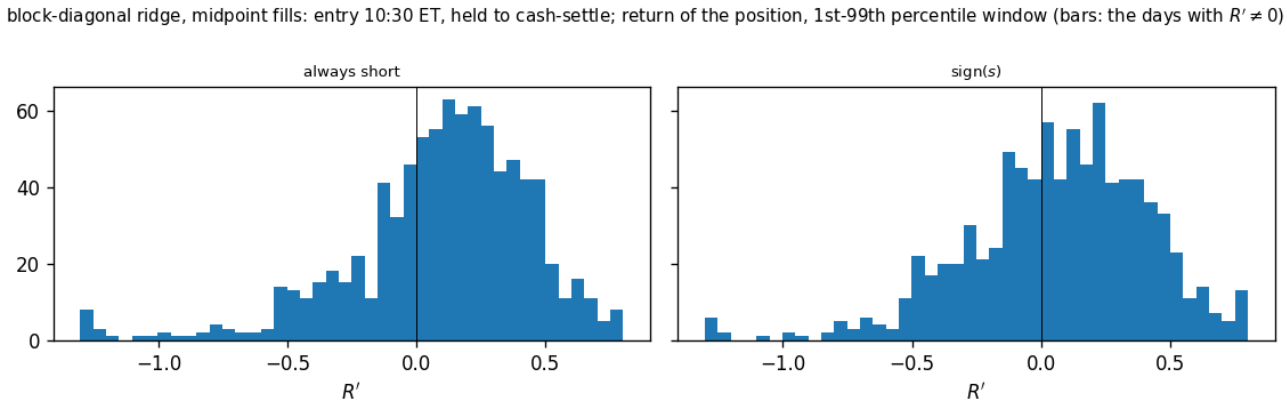


Distribution of the return of the position taken at 10:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 10:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 864 expiration days (the other 2 of the deck's 866 days have no quoted straddle at this stamp), 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 10:30 ET, held to cash-settle

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	865	0.096	0.366	-2.00	0.88	-1.44	4.42	7.74	4.18	0	0.0	3.51
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	865	0.055	0.375	-1.81	2.00	-0.39	2.87	4.35	2.35	199	23.0	1.70
block-diagonal ridge	865	0.066	0.373	-2.00	1.91	-0.65	3.15	5.19	2.80	210	24.3	2.15
block-diagonal ridge, without the FOMC columns	865	0.066	0.373	-2.00	1.91	-0.64	3.15	5.23	2.82	207	23.9	2.17
LightGBM†	865	0.066	0.373	-1.81	2.00	-0.38	2.96	5.20	2.81	197	22.8	2.15
XGBoost†	865	0.069	0.372	-1.81	2.00	-0.39	3.00	5.46	2.95	193	22.3	2.29
lasso (causally tuned)†	865	0.063	0.373	-2.00	1.91	-0.72	3.16	4.95	2.67	201	23.2	2.02
lasso ($\alpha = 10^{-4}$)	865	0.064	0.373	-2.00	1.91	-0.64	3.12	5.04	2.72	201	23.2	2.06
elastic net (causally tuned)†	865	0.074	0.372	-2.00	1.91	-0.64	3.24	5.82	3.14	198	22.9	2.48

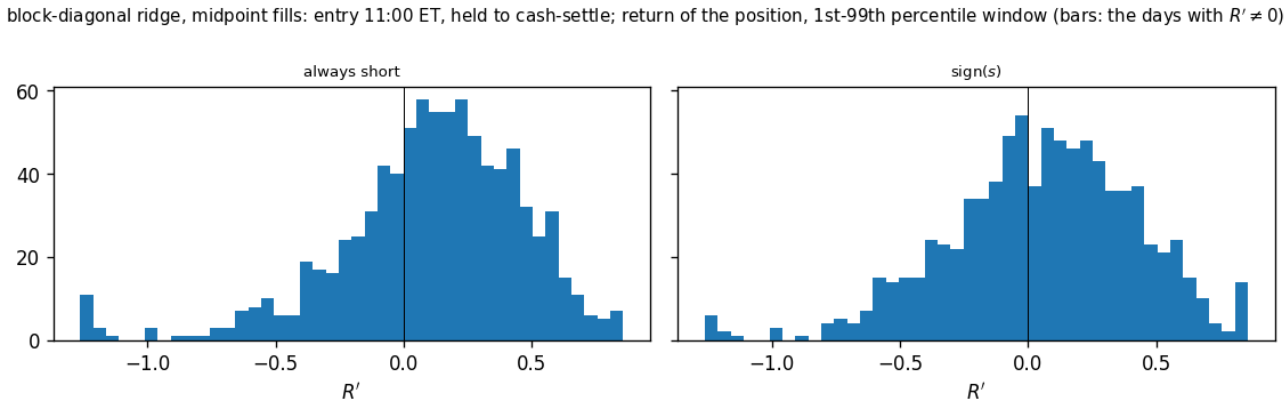


Distribution of the return of the position taken at 10:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 10:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 865 expiration days (the other 1 of the deck's 866 days have no quoted straddle at this stamp), 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 11:00 ET, held to cash-settle

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	865	0.103	0.388	-2.30	0.87	-1.57	5.53	7.78	4.20	0	0.0	3.58
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	865	0.057	0.398	-2.09	2.30	0.03	3.48	4.23	2.28	269	31.1	1.68
block-diagonal ridge	865	0.057	0.398	-2.09	2.30	-0.18	3.60	4.22	2.28	276	31.9	1.68
block-diagonal ridge, without the FOMC columns	865	0.070	0.396	-2.09	2.30	-0.13	3.66	5.17	2.79	275	31.8	2.18
LightGBM†	865	0.059	0.398	-2.09	2.30	-0.05	3.54	4.37	2.36	262	30.3	1.75
XGBoost†	865	0.064	0.397	-2.09	2.30	0.01	3.53	4.72	2.55	255	29.5	1.94
lasso (causally tuned)†	865	0.064	0.397	-2.09	2.30	-0.10	3.60	4.76	2.57	273	31.6	1.96
lasso ($\alpha = 10^{-4}$)	865	0.063	0.397	-2.09	2.30	-0.14	3.62	4.64	2.50	270	31.2	1.90
elastic net (causally tuned)†	865	0.059	0.397	-2.09	2.30	-0.15	3.60	4.39	2.37	273	31.6	1.77

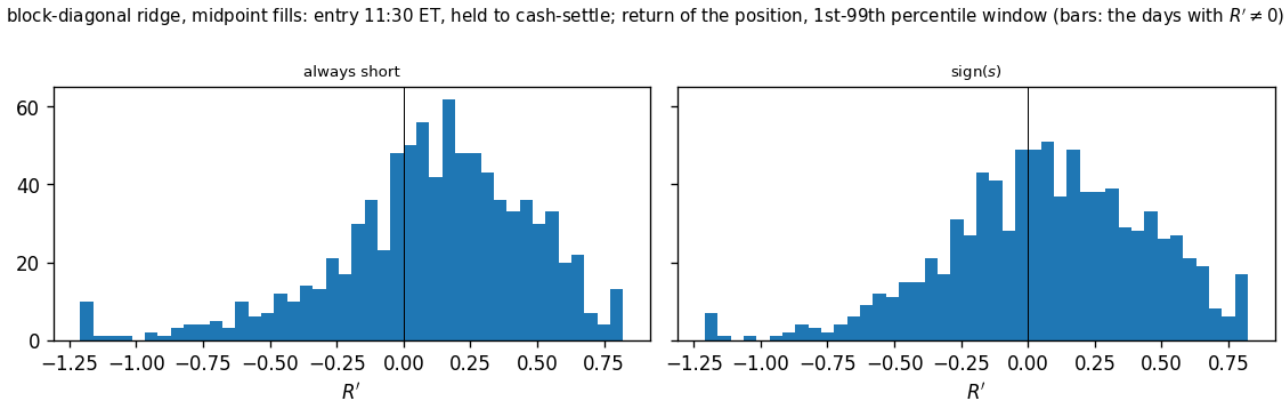


Distribution of the return of the position taken at 11:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 11:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 865 expiration days (the other 1 of the deck's 866 days have no quoted straddle at this stamp), 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 11:30 ET, held to cash-settle

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	865	0.101	0.407	-3.05	0.89	-1.81	8.19	7.32	3.95	0	0.0	3.33
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	865	0.064	0.414	-2.31	3.05	-0.25	6.08	4.53	2.45	231	26.7	1.84
block-diagonal ridge	865	0.074	0.412	-2.31	3.05	-0.09	6.09	5.30	2.86	252	29.1	2.25
block-diagonal ridge, without the FOMC columns	865	0.069	0.413	-2.31	3.05	-0.07	6.02	4.94	2.67	265	30.6	2.06
LightGBM†	865	0.058	0.415	-2.31	3.05	-0.28	6.03	4.10	2.21	221	25.5	1.60
XGBoost†	865	0.063	0.414	-2.31	3.05	-0.29	6.10	4.50	2.43	213	24.6	1.82
lasso (causally tuned)†	865	0.072	0.413	-2.31	3.05	-0.29	6.21	5.15	2.78	238	27.5	2.17
lasso ($\alpha = 10^{-4}$)	865	0.076	0.412	-2.31	3.05	-0.09	6.11	5.42	2.92	271	31.3	2.32
elastic net (causally tuned)†	865	0.075	0.412	-2.31	3.05	-0.23	6.20	5.34	2.88	236	27.3	2.27

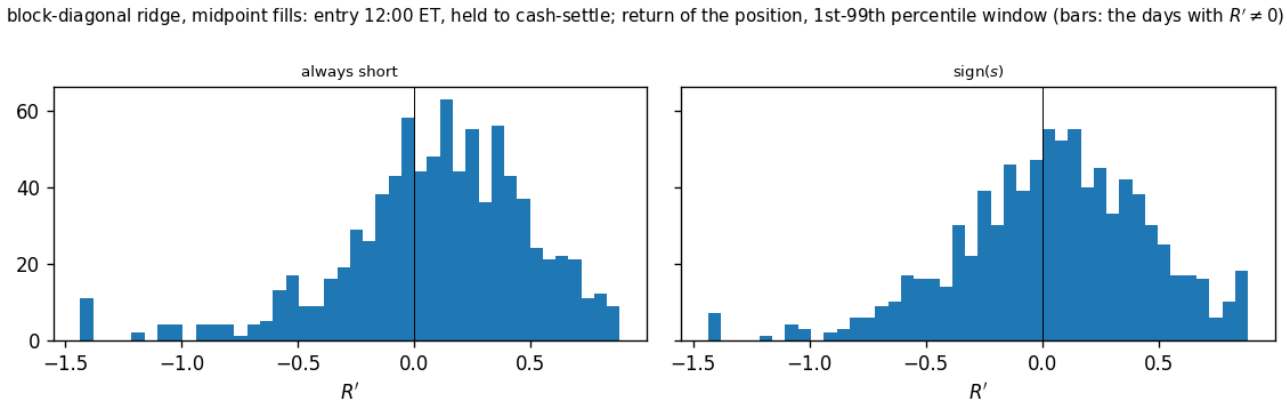


Distribution of the return of the position taken at 11:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 11:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 865 expiration days (the other 1 of the deck's 866 days have no quoted straddle at this stamp), 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 12:00 ET, held to cash-settle

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.087	0.434	-2.97	0.93	-1.48	5.45	5.88	3.17	0	0.0	2.58
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.032	0.442	-2.97	1.53	-0.90	4.25	2.16	1.17	259	29.9	0.58
block-diagonal ridge	866	0.048	0.440	-2.10	2.97	-0.29	4.16	3.21	1.73	269	31.1	1.15
block-diagonal ridge, without the FOMC columns	866	0.046	0.440	-2.10	2.97	-0.28	4.14	3.06	1.65	275	31.8	1.07
LightGBM†	866	0.054	0.440	-2.10	2.97	-0.33	4.23	3.63	1.96	246	28.4	1.38
XGBoost†	866	0.043	0.441	-2.10	2.97	-0.32	4.14	2.89	1.56	251	29.0	0.98
lasso (causally tuned)†	866	0.043	0.441	-2.97	1.53	-0.90	4.37	2.90	1.57	265	30.6	0.98
lasso ($\alpha = 10^{-4}$)	866	0.045	0.441	-2.10	2.97	-0.28	4.14	3.00	1.62	266	30.7	1.04
elastic net (causally tuned)†	866	0.051	0.440	-2.97	1.53	-0.91	4.47	3.41	1.84	265	30.6	1.26

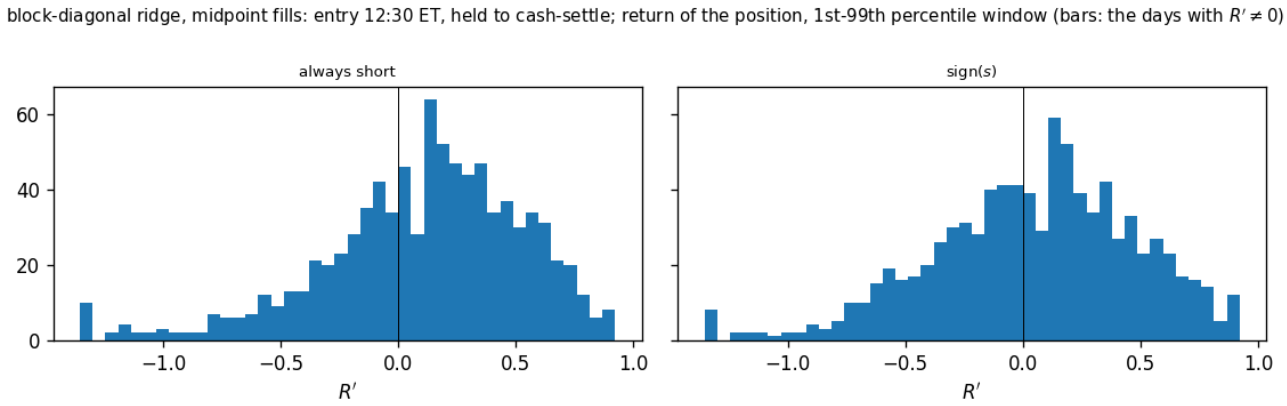


Distribution of the return of the position taken at 12:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 12:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 12:30 ET, held to cash-settle

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.097	0.466	-4.31	0.93	-1.88	10.76	6.12	3.30	0	0.0	2.72
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.058	0.472	-2.37	4.31	0.33	8.52	3.58	1.93	252	29.1	1.36
block-diagonal ridge	866	0.050	0.473	-2.37	4.31	0.30	8.49	3.13	1.69	278	32.1	1.12
block-diagonal ridge, without the FOMC columns	866	0.050	0.473	-2.37	4.31	0.29	8.50	3.12	1.69	285	32.9	1.11
LightGBM†	866	0.037	0.474	-2.37	4.31	0.32	8.43	2.32	1.25	268	30.9	0.68
XGBoost†	866	0.054	0.473	-2.37	4.31	0.34	8.50	3.37	1.82	259	29.9	1.24
lasso (causally tuned)†	866	0.062	0.472	-2.37	4.31	0.41	8.51	3.87	2.09	269	31.1	1.51
lasso ($\alpha = 10^{-4}$)	866	0.052	0.473	-2.37	4.31	0.31	8.50	3.23	1.74	277	32.0	1.17
elastic net (causally tuned)†	866	0.055	0.472	-2.37	4.31	0.33	8.50	3.41	1.84	262	30.3	1.27

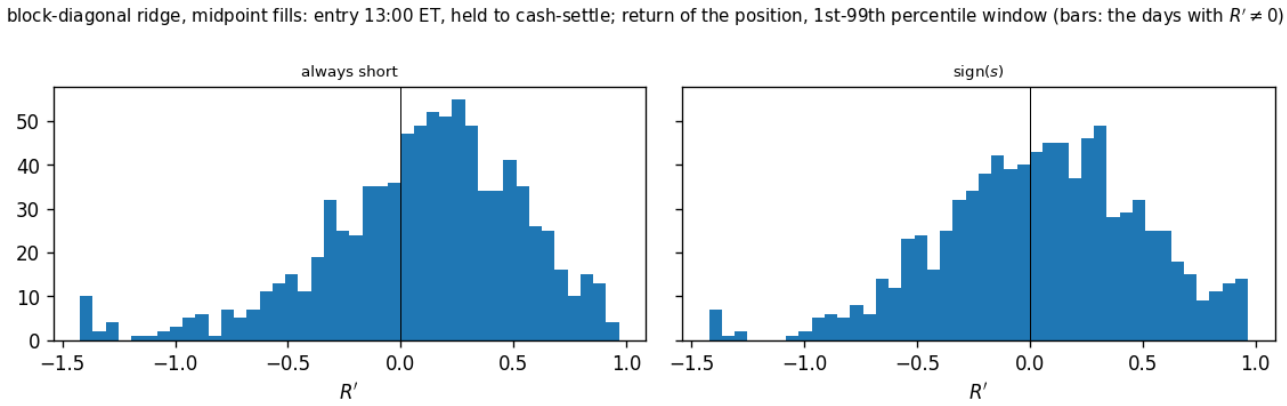


Distribution of the return of the position taken at 12:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 12:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 13:00 ET, held to cash-settle

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.083	0.498	-5.47	0.95	-2.48	19.28	4.93	2.66	0	0.0	2.09
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.042	0.503	-2.44	5.47	1.06	16.54	2.47	1.33	335	38.7	0.77
block-diagonal ridge	866	0.053	0.502	-2.44	5.47	1.05	16.60	3.12	1.68	334	38.6	1.12
block-diagonal ridge, without the FOMC columns	866	0.051	0.502	-2.44	5.47	1.04	16.59	2.97	1.60	341	39.4	1.04
LightGBM†	866	0.061	0.501	-2.44	5.47	1.20	16.58	3.60	1.94	341	39.4	1.38
XGBoost†	866	0.053	0.502	-2.44	5.47	1.08	16.58	3.09	1.67	339	39.1	1.10
lasso (causally tuned)†	866	0.055	0.502	-2.44	5.47	1.04	16.61	3.21	1.73	323	37.3	1.17
lasso ($\alpha = 10^{-4}$)	866	0.053	0.502	-2.44	5.47	1.05	16.59	3.11	1.68	330	38.1	1.11
elastic net (causally tuned)†	866	0.056	0.501	-2.44	5.47	1.04	16.62	3.31	1.78	319	36.8	1.22

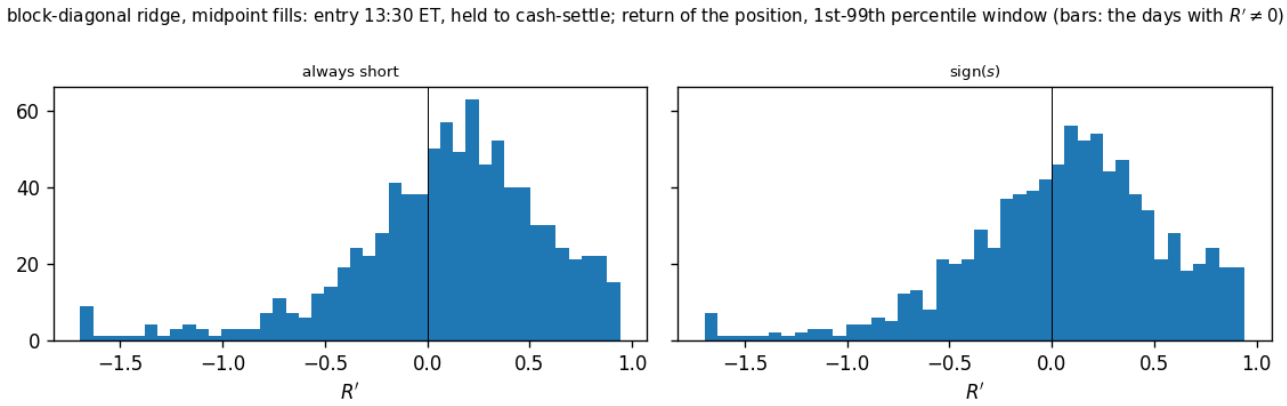


Distribution of the return of the position taken at 13:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 13:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 13:30 ET, held to cash-settle

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.088	0.557	-6.72	1.00	-3.24	28.71	4.65	2.51	0	0.0	1.97
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.067	0.560	-3.14	6.72	1.21	25.51	3.51	1.90	189	21.8	1.36
block-diagonal ridge	866	0.073	0.559	-3.14	6.72	1.25	25.58	3.83	2.06	205	23.7	1.53
block-diagonal ridge, without the FOMC columns	866	0.074	0.559	-3.14	6.72	1.26	25.59	3.88	2.09	215	24.8	1.56
LightGBM†	866	0.055	0.561	-3.14	6.72	1.22	25.37	2.87	1.55	193	22.3	1.01
XGBoost†	866	0.065	0.560	-3.14	6.72	1.22	25.48	3.42	1.84	197	22.7	1.31
lasso (causally tuned)†	866	0.059	0.561	-3.14	6.72	1.26	25.39	3.08	1.66	200	23.1	1.13
lasso ($\alpha = 10^{-4}$)	866	0.071	0.559	-3.14	6.72	1.24	25.56	3.75	2.02	205	23.7	1.49
elastic net (causally tuned)†	866	0.052	0.561	-3.14	6.72	1.28	25.33	2.73	1.47	194	22.4	0.94

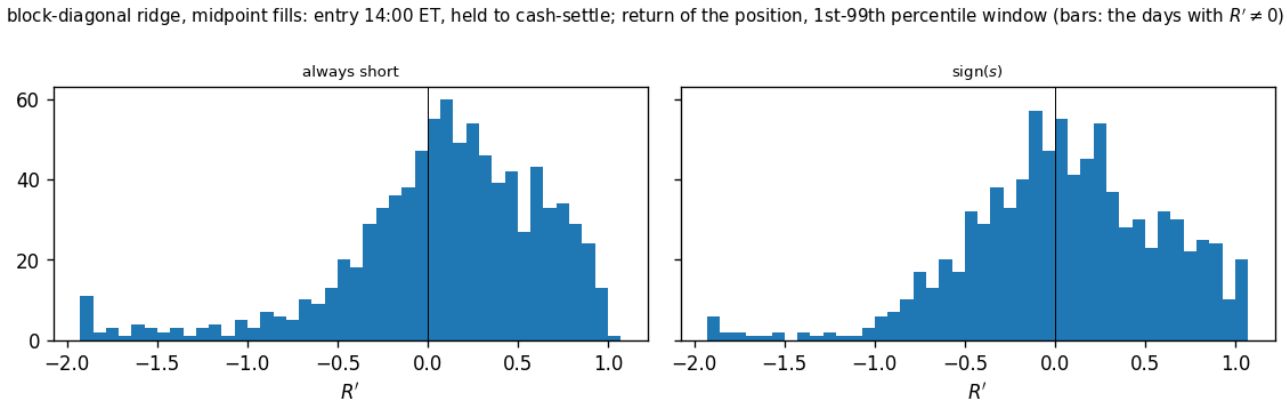


Distribution of the return of the position taken at 13:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 13:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 14:00 ET, held to cash-settle

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.080	0.623	-6.94	1.00	-2.74	20.45	3.77	2.04	0	0.0	1.36
sign(s) (± 1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.054	0.626	-2.93	6.94	1.17	18.30	2.52	1.36	335	38.7	0.69
block-diagonal ridge	866	0.066	0.625	-2.93	6.94	1.12	18.37	3.09	1.67	325	37.5	1.00
block-diagonal ridge, without the FOMC columns	866	0.073	0.624	-2.93	6.94	1.17	18.39	3.44	1.86	333	38.5	1.19
LightGBM†	866	0.055	0.626	-2.93	6.94	1.16	18.30	2.57	1.39	350	40.4	0.72
XGBoost†	866	0.053	0.626	-2.93	6.94	1.18	18.30	2.51	1.36	360	41.6	0.69
lasso (causally tuned)†	866	0.061	0.625	-2.93	6.94	1.15	18.33	2.86	1.54	337	38.9	0.87
lasso ($\alpha = 10^{-4}$)	866	0.068	0.624	-2.93	6.94	1.16	18.37	3.23	1.74	325	37.5	1.07
elastic net (causally tuned)†	866	0.049	0.626	-2.93	6.94	1.09	18.31	2.30	1.24	334	38.6	0.58

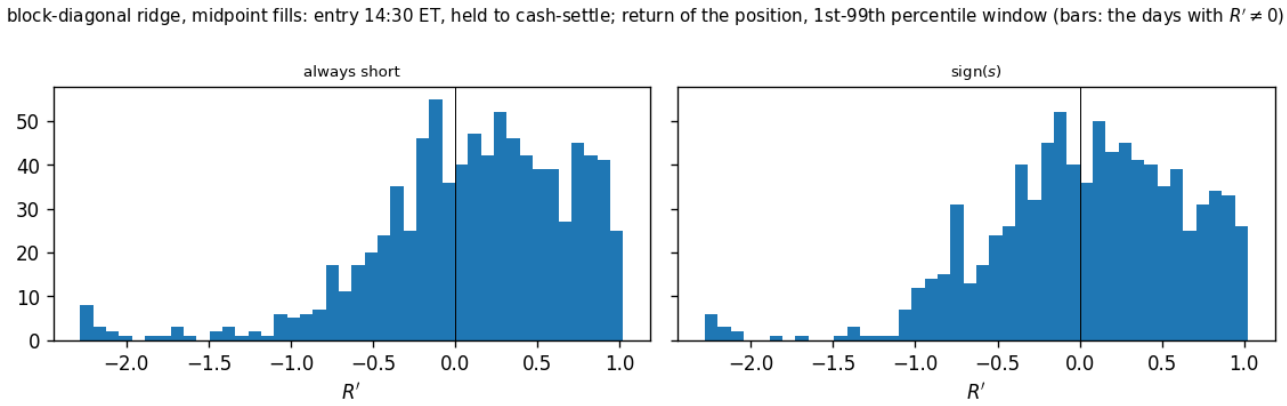


Distribution of the return of the position taken at 14:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 14:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 14:30 ET, held to cash-settle

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.091	0.718	-8.43	1.05	-3.71	32.65	3.71	2.00	0	0.0	1.50
sign(s) (*± 1 on the sign of s*)												
baseline (HAR + calendar OLS)	866	0.045	0.722	-6.46	8.43	0.35	29.96	1.84	0.99	232	26.8	0.49
block-diagonal ridge	866	0.037	0.722	-8.43	2.98	-3.15	30.61	1.50	0.81	243	28.1	0.31
block-diagonal ridge, without the FOMC columns	866	0.030	0.723	-8.43	2.98	-3.13	30.43	1.21	0.65	250	28.9	0.15
LightGBM†	866	0.028	0.723	-8.43	2.21	-3.25	30.40	1.13	0.61	263	30.4	0.11
XGBoost†	866	0.031	0.723	-8.43	2.98	-3.12	30.47	1.27	0.69	263	30.4	0.18
lasso (causally tuned)†	866	0.043	0.722	-6.46	8.43	0.40	29.93	1.76	0.95	252	29.1	0.45
lasso ($\alpha = 10^{-4}$)	866	0.042	0.722	-8.43	2.98	-3.14	30.75	1.71	0.92	246	28.4	0.42
elastic net (causally tuned)†	866	0.046	0.722	-6.46	8.43	0.36	29.96	1.88	1.01	250	28.9	0.51

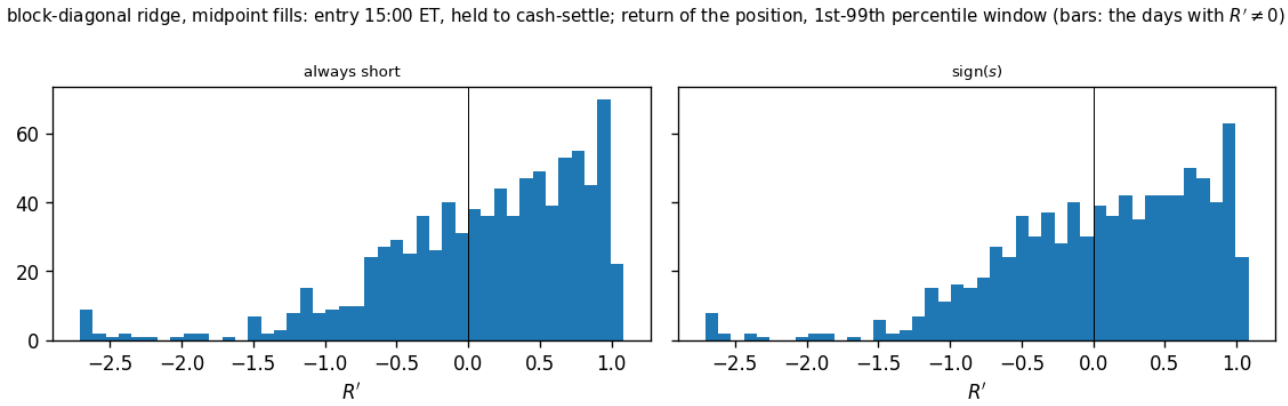


Distribution of the return of the position taken at 14:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 14:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 15:00 ET, held to cash-settle

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.095	0.778	-5.81	1.04	-1.95	7.83	3.59	1.93	0	0.0	1.40
sign(s) (± 1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.064	0.781	-5.81	3.95	-1.44	7.24	2.41	1.30	105	12.1	0.76
block-diagonal ridge	866	0.060	0.782	-5.81	3.95	-1.43	7.20	2.26	1.22	108	12.5	0.69
block-diagonal ridge, without the FOMC columns	866	0.058	0.782	-5.81	3.95	-1.43	7.17	2.17	1.17	110	12.7	0.64
LightGBM†	866	0.068	0.781	-5.81	3.95	-1.45	7.29	2.58	1.39	110	12.7	0.86
XGBoost†	866	0.074	0.780	-5.81	3.95	-1.39	7.33	2.79	1.51	115	13.3	0.97
lasso (causally tuned)†	866	0.067	0.781	-5.81	3.95	-1.47	7.28	2.51	1.35	111	12.8	0.82
lasso ($\alpha = 10^{-4}$)	866	0.046	0.782	-5.81	2.45	-1.72	7.13	1.74	0.94	106	12.2	0.40
elastic net (causally tuned)†	866	0.061	0.781	-5.81	3.95	-1.45	7.22	2.31	1.25	115	13.3	0.71

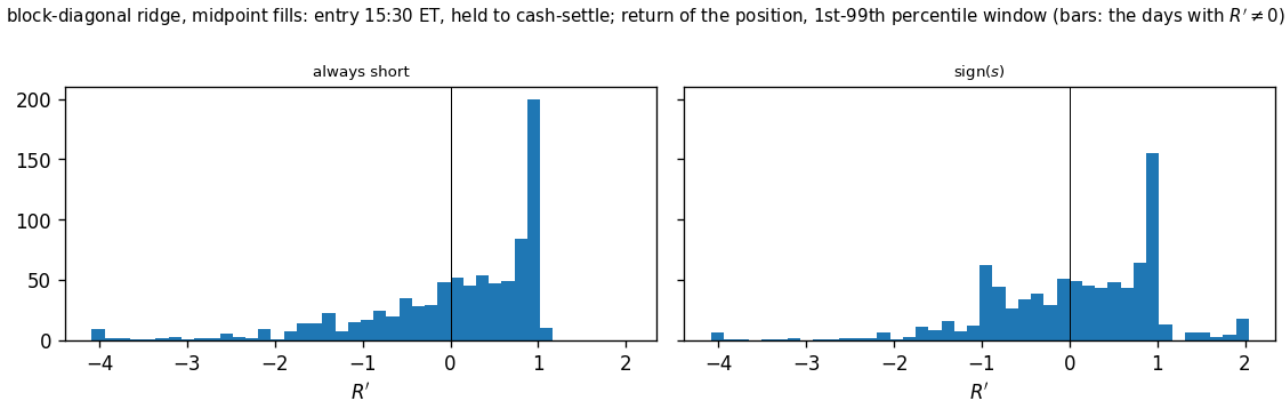


Distribution of the return of the position taken at 15:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 15:00 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — entry 15:30 ET, held to cash-settle

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.017	1.115	-9.38	1.08	-2.26	9.18	0.44	0.24	0	0.0	-0.23
sign(s) (± 1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.027	1.115	-9.38	5.49	-1.47	9.19	0.71	0.39	148	17.1	-0.09
block-diagonal ridge	866	0.081	1.113	-6.37	9.38	0.05	9.12	2.14	1.15	234	27.0	0.68
block-diagonal ridge, without the FOMC columns	866	0.082	1.113	-6.37	9.38	0.04	9.12	2.17	1.17	236	27.3	0.70
LightGBM†	866	0.103	1.111	-6.37	9.38	0.02	9.19	2.74	1.48	165	19.1	1.01
XGBoost†	866	0.091	1.112	-9.38	5.49	-1.34	9.60	2.40	1.30	162	18.7	0.82
lasso (causally tuned)†	866	0.102	1.111	-6.37	9.38	-0.15	9.25	2.71	1.46	184	21.2	0.99
lasso ($\alpha = 10^{-4}$)	866	0.064	1.114	-6.37	9.38	-0.22	9.15	1.70	0.91	246	28.4	0.44
elastic net (causally tuned)†	866	0.062	1.114	-6.37	9.38	-0.11	9.12	1.65	0.89	182	21.0	0.42

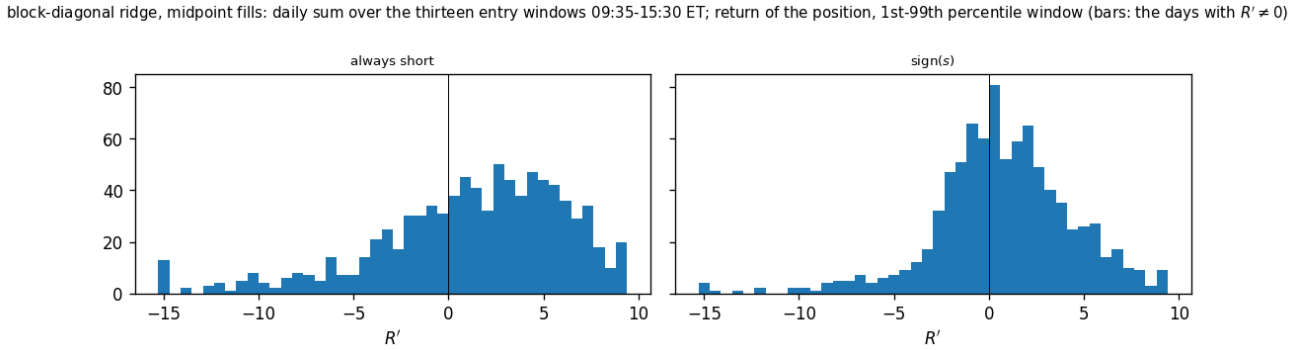


Distribution of the return of the position taken at 15:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window; the mass at -1 is the days the straddle expires worthless).

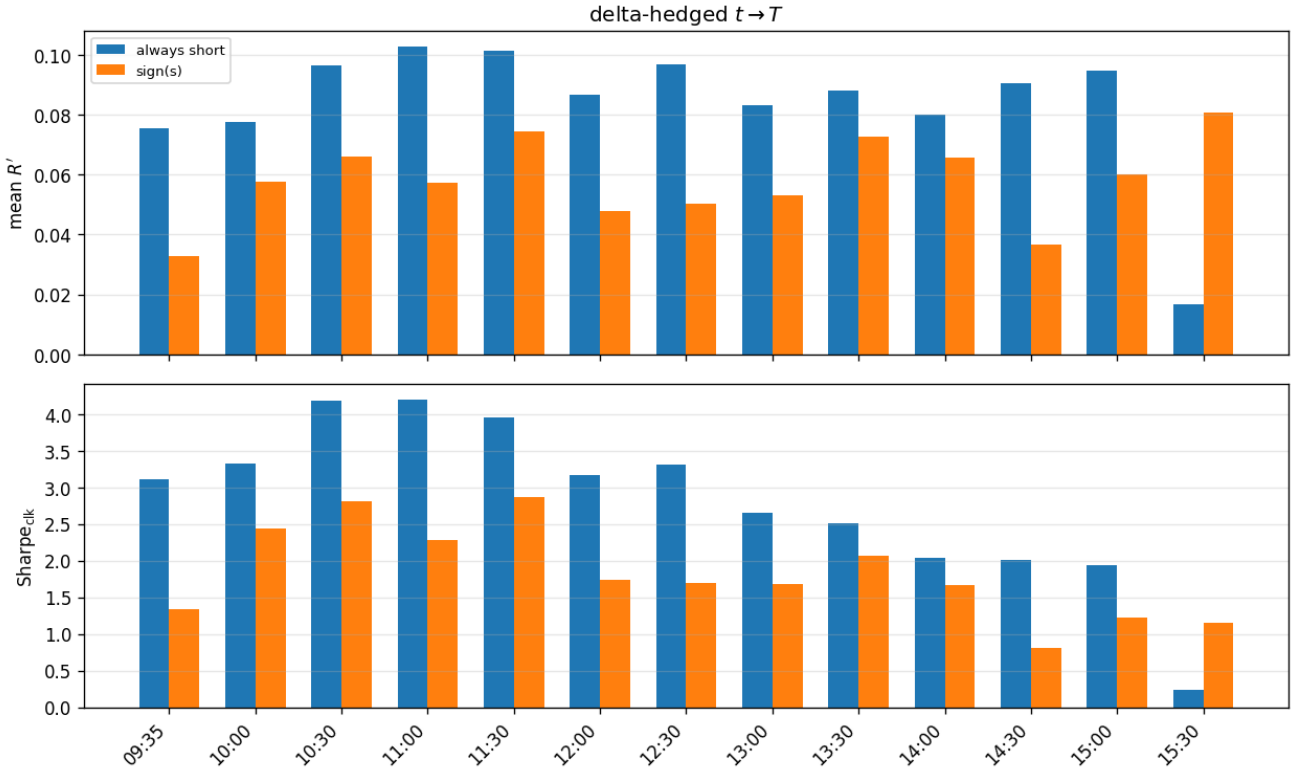
One trade a day: enter the nearest-OTM straddle at 15:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Rule table by strategy — pooled over the thirteen entry windows

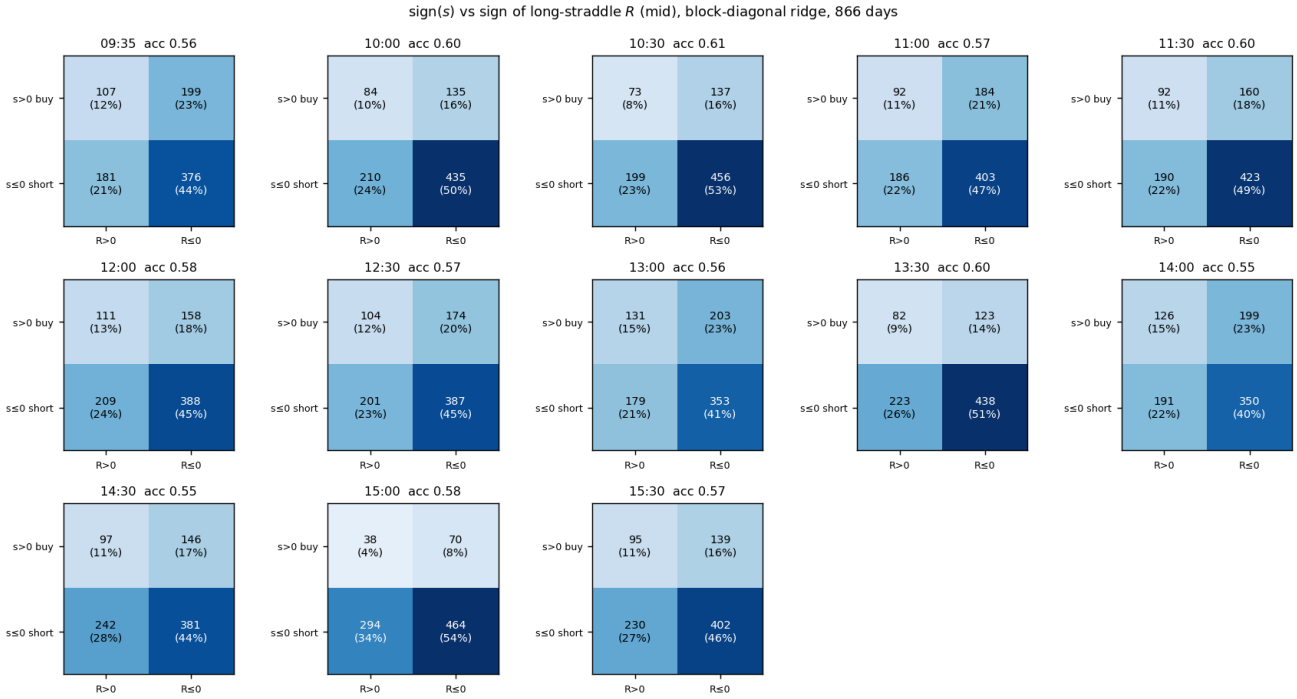
	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{ann}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{ann}}$
Short volatility (always short)												
all models (no forecast)	866	1.090	5.746	-58.68	10.56	-2.51	16.34	5.58	3.01	0	0.0	2.28
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.660	3.896	-26.83	33.98	-0.17	9.67	4.99	2.69	143	16.5	1.61
block-diagonal ridge	866	0.755	4.053	-26.83	41.81	0.19	16.37	5.48	2.96	179	20.7	1.92
block-diagonal ridge, without the FOMC columns	866	0.755	4.022	-26.83	41.81	0.24	16.65	5.52	2.98	192	22.2	1.94
LightGBM†	866	0.752	3.936	-26.83	41.81	0.41	17.58	5.63	3.03	147	17.0	1.97
XGBoost†	866	0.764	3.729	-26.83	23.05	-0.77	6.81	6.03	3.25	148	17.1	2.13
lasso (causally tuned)†	866	0.783	4.152	-26.83	52.74	1.34	32.20	5.55	3.00	159	18.4	1.98
lasso ($\alpha = 10^{-4}$)	866	0.730	3.983	-26.83	33.92	-0.40	10.24	5.39	2.91	187	21.6	1.86
elastic net (causally tuned)†	866	0.728	4.214	-26.83	52.74	1.19	30.81	5.08	2.74	162	18.7	1.74



Distribution of the return of the daily sum over the thirteen entry windows, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).



Mean R' and $\text{Sharpe}_{\text{clk}}$ by entry clock (grouped bars). $\text{Sharpe}_{\text{clk}}$ is that clock's daily series $\times\sqrt{252}$, not the pooled $\text{Sharpe}_{\text{ann}}$. Delta-hedged $t \to T$: hold entry K to official close, rebalance Δ every 30 minutes (from 09:35 the first rebalance is at 10:00).



Confusion matrix per entry clock: sign(s) (buy if $s > 0$) against the sign of the long-straddle midpoint return. 15:30 is the last-30-min trade.

Each row is the daily sum of the thirteen overlapping remaining-session books: enter at each clock 09:35–15:30 ET, hold those strikes to the official close, delta-hedge every 30 minutes (from 09:35 the first re-hedge is at 10:00). 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread).

Notes

Straddle here means the nearest out-of-the-money call plus the nearest out-of-the-money put, same-day expiry, bought or sold as one position. One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean/std}$, with no autocorrelation correction. "ex. kurt" is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. **Two Sharpes.** Both multiply by $\sqrt{252}$ because an expiration day is the unit of time (same year-length as the paper). They are not the same object. On a *single-clock* page, $\text{Sharpe}_{\text{clk}} = (\text{mean/std}) \times \sqrt{252}$ of that clock's series: one remaining-session short per expiration day (enter at this clock, hold those strikes to the close, delta-hedge every 30 minutes). It answers "if I only entered at this clock, what is my annual Sharpe?" It is *not* the book's $\text{Sharpe}_{\text{ann}}$. On the *pooled* page, $\text{Sharpe}_{\text{ann}} = (\text{mean/std}) \times \sqrt{252}$ of $R_d^{day} = \sum_t R'_{d,t}$, the sum of the day's thirteen overlapping remaining-session books. That reserved name is the book's daily P&L after pooling the session. The thirteen books overlap; they are not thirteen successive holds. $\text{Sharpe}^{\times}$ is the same convention at the crossed spread. Every other column is daily. n_{buy} is the number of expiration days with position > 0 (always-short is 0; on the pooled page it is days the sum of the thirteen positions is positive); %buy is $100 \cdot n_{\text{buy}}/n$, the notebook's column. The other column this table adds to the deck's is $\text{Sharpe}^{\times}_{\text{ann}}$, the same rule's annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck's.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - \text{IV}^2_{\text{hr}} h_t w_t$: the forecast against the implied variance of the same window. w_t is the expanding mean of each prior day's remaining-session share of realized variance, $\text{RV}_t / \sum_{s \geq t} \text{RV}_s$ (in-fit panel history back to 2001), not the ratio of trailing clock-means. On the bars whose vendor implied volatility is a censored solver node the deck's treatment is used here too — the volatility that reproduces the straddle midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat. At the 09:35 entry the forecast is the one issued at 09:30 for the 09:30–10:00 bar, w_t is that bar's share, and h_t is the chain's hours to the close.